

2. Analice y justifique la convergencia de la siguiente serie, y calcule su suma

$$\sum_{k=3}^{\infty} \frac{9 \cdot 5^{-k}}{(-3)^{2-k}} - \frac{2}{k^2 - 1}$$

$\infty$

$$\sum_{k=3}^{\infty} \frac{9 \cdot 5^{-k}}{(-3)^2 \cdot (-3)^{-k}}$$

$\infty$

$$\sum_{k=3}^{\infty} \frac{2}{k^2 - 1}$$

$\infty$

$$\sum_{k=3}^{\infty} \left( \frac{5^{-1}}{(-3)^{-1}} \right)^k$$

$\infty$

$$\sum_{k=3}^{\infty} \frac{2}{(k-1)(k+1)}$$

$\infty$

$$\sum_{k=3}^{\infty} \left( \frac{-3}{5} \right)^k$$

$$\frac{2}{(k-1)(k+1)} = \frac{A}{k-1} + \frac{B}{k+1}$$

$$\left| \frac{-3}{5} \right| = \frac{3}{5} < 1$$

$$k-1=0 \quad k=1$$

$$k+1=0 \quad k=-1$$

$$\frac{\left( \frac{-3}{5} \right)^3}{1 - \frac{3}{5}}$$

$$2 = A(k+1) + B(k-1)$$

$$k=1 \rightarrow 2 = 2A$$

$$A=1$$

$$\frac{-27}{200}$$

$$k=-1 \quad 2 = -2B$$

$$B=-1$$

$$\frac{2}{(k-1)(k+1)} = \frac{1}{k-1} - \frac{1}{k+1}$$

$$a_n - a_{n+1}$$

$$\infty \quad \varepsilon \quad \frac{1}{n-1} - \frac{1}{n+1} \quad \frac{(n+1)-1}{n}$$

$$\infty \quad \varepsilon \quad \left( \frac{1}{n-1} - \frac{1}{n} \right) + \varepsilon \quad \left( \frac{1}{n} - \frac{1}{n+1} \right)$$

$$\frac{1}{3-1} - \lim_{n \rightarrow \infty} \frac{1}{n} + \frac{1}{3} - \lim_{n \rightarrow \infty} \frac{1}{n+1}$$

$$\frac{1}{2} + \frac{1}{3} = \frac{5}{6}$$

$$\frac{-27}{200} + \frac{5}{6}$$

|     |
|-----|
| 439 |
| 600 |

$$a_n - a_{n+1}$$

$$\frac{1}{n-1} - \frac{1}{n+1}$$

$$\left( \frac{1}{n-1} - \frac{1}{n} \right) + \left( \frac{1}{n} - \frac{1}{n+1} \right)$$

$$a_n - a_{n+1}$$

$$n - n+1 \quad \checkmark$$

$$n - n+2$$

$$\hookrightarrow n+1$$

$$(n - n+1) + (n+1 - n+2)$$

$$n+1 \rightarrow$$

$$n+1+1 \rightarrow$$

$$1. \sum_{n=2}^{\infty} \frac{(-1)^n \cdot (n^2 - n) \cdot 3^n + 5^n}{5^n \cdot (n^2 - n)}$$

$\infty$

$$\sum_{n=2}^{\infty} \frac{(-1)^n \cancel{(n^2 - n)} \cdot 3^n}{5^n \cancel{(n^2 - n)}} + \frac{\cancel{5^n}}{\cancel{5^n \cancel{(n^2 - n)}}}$$

$\infty$

$$\sum_{n=2}^{\infty} \left( \frac{-1 \cdot 3}{5} \right)^n + \sum_{n=2}^{\infty} \frac{1}{n(n-1)}$$

$$\frac{\left(\frac{-3}{5}\right)^2}{1 - \frac{-3}{5}}$$

$$\frac{1}{n(n-1)} = \frac{A}{n-1} + \frac{B}{n}$$

$n=1, n=0$

$$1 = A(n) + B(n-1)$$

|    |
|----|
| 9  |
| 70 |

$$n=1 \rightarrow 1 = A$$

$$A=1$$

$$n=0 \rightarrow 1 = -B$$

$$B = -1$$

$$\sum_{n=2}^{\infty} \frac{1}{n-1} - \frac{1}{n}$$

$$\frac{1}{2-1} - \lim_{n \rightarrow \infty} \frac{1}{n}$$

$$\frac{9}{70} + 1$$

$\Downarrow$

|     |
|-----|
| .99 |
| 70  |

$$3. \sum_{m=2}^{\infty} \frac{1}{2^{2m+1}} - \frac{1}{m} + \frac{1}{m+1}$$

$$\sum_{n=2}^{\infty} \frac{1}{2^{2n+1}} + \sum_{n=2}^{\infty} \frac{-1}{n} + \frac{1}{n+1}$$

$$\sum_{n=2}^{\infty} \frac{1}{(2^2)^n \cdot 2} - \sum_{n=2}^{\infty} \frac{1}{n} - \frac{1}{n+1}$$

$$\frac{1}{2} \sum_{n=2}^{\infty} \left(\frac{1}{4}\right)^n - \sum_{n=2}^{\infty} \frac{1}{n} - \frac{1}{n+1}$$

$$\frac{\frac{1}{2} \cdot \left(\frac{1}{4}\right)^2}{1 - \frac{1}{4}} - \frac{1}{2} - \lim_{n \rightarrow \infty} \frac{1}{n+1}$$

|                  |
|------------------|
| $\frac{-11}{24}$ |
|------------------|

Series alternadas

$$\sum_{n=p}^{\infty} a_n \underbrace{(-1)^n}_{\substack{\rightarrow (-1)^{n-1} \quad (-1)^{n+1} \dots}}$$

Para convergen

1)  $a_n$  sea decreciente

$$\begin{cases} f'(x) \\ \frac{a_{n+1}}{a_n} \geq 1 \end{cases} \begin{cases} \checkmark \rightarrow \text{cruce} \\ \times \rightarrow \text{Decrece} \end{cases}$$

2)  $\lim_{n \rightarrow \infty} a_n = 0$

$$\sum_{n=p}^{\infty} a_n \underbrace{(-1)^n}_{\text{red}} \rightarrow \sum_{n=p}^{\infty} a_n \quad \begin{cases} \text{Si ya converge,} \\ \text{Convergencia absoluta} \end{cases}$$

$$\sum_{n=p}^{\infty} a_n \underbrace{(-1)^n}_{\text{red}} \rightarrow \sum_{n=p}^{\infty} a_n \quad \begin{cases} \text{Diverge, aplicamos} \\ \text{criterio de las} \\ \text{series alternadas} \end{cases}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} (-1)^n \rightarrow \frac{1}{n^2} \quad \text{p serie, } p > 1$$

∴ Converge

∴ Converge absolutamente

$$\sum_{n=1}^{\infty} \frac{1}{n} (-1)^n \rightarrow \frac{1}{n} \quad \text{p serie, } p \leq 1$$

∴ Diverge

Decreciente

$$f'(x) = -\frac{1}{x^2} \quad \text{Decreciente} \quad \checkmark$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0 \quad \checkmark$$

∴ La serie converge condicionalmente

$\infty$ 

$$\sum_{n=1}^{\infty} \frac{1}{n \ln(n)} (-1)^{n-1}$$

 $\infty$ 

$$\sum_{n=1}^{\infty} \frac{1}{n \ln(n)}$$

Por el criterio de la integral

$$f(x) = \frac{1}{x \ln(x)}$$

$$-1, f'(x) \cdot g(x) + f(x) \cdot g'(x)$$

$$f'(x) = \frac{-1}{(x \ln(x))^2} \cdot \ln(x) + \cancel{x \cdot \frac{1}{x}} \left[ \frac{1}{f(x)} \right]' = \frac{-1}{f^2(x)} f'(x)$$

$$[f(x)g(x)]' = f'(x)g(x) + f(x)g'(x)$$

$$= - \frac{\ln(x) + 1}{(x \ln(x))^2}$$

$$x + \cancel{x} = x$$

Decreciente

$$\int_1^{+\infty} \frac{1}{x \ln(x)} dx \longrightarrow \int \frac{1}{x \ln(x)} dx \quad \begin{matrix} u = \ln(x) \\ du = \frac{1}{x} dx \end{matrix}$$

$$\int \frac{1}{u} du$$

$$\begin{matrix} \ln|u| & +\infty \\ \ln|\ln|x|| & | \\ & 1 \end{matrix}$$

$$\ln(+\infty) - \ln|\ln(1)|$$

$+\infty - 0 = +\infty$  i.e. Divergente

En el paso se verificó que era decreciente

$$\lim_{n \rightarrow +\infty} \frac{1}{n \ln(n)} = 0$$

$$\sum_{n=1}^{\infty} \frac{1}{n \ln(n)} (-1)^{n-1}$$

Converge condicionalmente

p series

$\infty$

$$\sum_{n=p}^{\infty} \frac{1}{n^p}, \quad p > 1, \text{ Converge}$$

$$p \leq 1, \text{ Diverge}$$

$$\frac{1}{n^{\frac{1}{2}}}, \quad p = \frac{1}{2} < 1, \text{ Diverge}$$

$$\frac{1}{n^7}, \quad p = 7 > 1, \text{ Converge}$$